

Event-TimeRAF: Event-Aware Retrieval-Augmented Foundation Models for Explainable Air Quality Forecasting Under Concept Drift

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Abstract—Urban air-quality forecasting is important for public-health warnings, environmental management, and city-scale decision support. Recent time-series foundation models have shown useful zero-shot forecasting capability, and TimeRAF enriches such models by retrieving relevant time-series sequences from a knowledge base and incorporating them into a frozen time-series foundation model. However, air-pollution variation is often influenced by outside context, including meteorology, holidays, wildfire smoke, high-wind events, heavy rain, and other environmental factors. This paper presents Event-TimeRAF, an event-aware retrieval-augmented forecasting framework for Los Angeles County PM2.5 prediction. The framework combines historical air-quality windows, weather features, calendar context, retrieved time-series patterns, source-recorded environmental events, distribution-shift indicators, and evidence-based explanation generation. We evaluate the method on official EPA AirData PM2.5 observations, NOAA Global Hourly weather, and hash-verified NOAA Storm Events records for 2019–2024. In the final Kaggle run, the best overall model is the weather-calendar XGBoost context baseline with MSE 26.185, while the full Event-TimeRAF XGBoost variant obtains MSE 26.712. The frozen Chronos-Bolt baseline obtains MSE 28.941, and retrieval-augmented Chronos improves it to MSE 28.709. Because NOAA Storm Events lacks machine-readable publication timestamps, event-aware results are reported as retrospective event-start sensitivity results rather than strict real-time event forecasts.

Index Terms—Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, TimeRAF, concept drift, explainable forecasting.

I. INTRODUCTION

Air pollution is a major urban risk factor, and fine particulate matter (PM2.5) is especially important because it is associated with respiratory and cardiovascular health effects [1]. Forecasting PM2.5 can support public-health alerts, exposure reduction, transport planning, and environmental policy. In Los Angeles County, air quality can change because of meteorological conditions, traffic patterns, seasonal behavior, industrial activity, wildfire smoke, and unusual environmental events. These properties make air-quality forecasting a difficult time-series forecasting task.

Classical forecasting models and machine-learning models such as ARIMA, tree-based regressors, recurrent neural networks, and Transformer-based models can learn historical patterns when training and testing conditions are similar. However, urban air quality is strongly context-dependent. A model trained only on past numerical observations may

perform poorly during unusual weather, public holidays, traffic disruptions, fire incidents, dust events, or other external shocks. This limitation is closely connected to concept drift, where the data-generating process changes over time.

Time-series foundation models (TSFMs) have recently been proposed to improve generalization across forecasting tasks. Models such as TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers (TTM) aim to transfer forecasting knowledge learned from large time-series corpora to new downstream domains [2]–[6]. TimeRAF takes this direction further by introducing retrieval augmentation for zero-shot time-series forecasting [7]. It builds a time-series knowledge base, retrieves relevant candidate sequences, and integrates the retrieved knowledge using Channel Prompting while keeping the TSFM backbone frozen.

TimeRAF is a strong base method, but the retrieved knowledge is mainly numerical time-series data. For air-quality forecasting, external context is important: rainfall can wash out particulates, low wind speed can increase stagnation, holidays can change traffic patterns, and wildfire or smoke events can produce abrupt changes. Therefore, a natural extension is to retrieve not only similar time-series patterns, but also event, weather, and calendar context relevant to the forecast horizon.

This paper presents Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Los Angeles County PM2.5 forecasting under distribution shift. The main contributions are:

- an event-aware retrieval-augmented forecasting framework for urban PM2.5 prediction;
- a hybrid knowledge-base design that combines air-quality windows, weather summaries, calendar information, and event/news records;
- a TimeRAF-style retrieval baseline using random retrieval and similarity-based retrieval over historical windows;
- simple concept-drift indicators based on distribution shift, retrieval similarity, weather changes, and event bursts;
- evidence-based explanation generation using retrieved historical cases, event context, weather drivers, drift indicators, and feature importance;
- an ablation study comparing no retrieval, random retrieval, time-series retrieval, event retrieval, drift-aware retrieval, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt.

The final experiment uses official-source records for 2019–2024 and reports only values generated by the saved Kaggle run artifacts. The results are mixed: the retrieval-augmented

Chronos model improves MSE over frozen Chronos-Bolt, but the strongest overall model is a conventional XGBoost weather-calendar context baseline. This outcome motivates a careful discussion of where retrieval helps and where the current feature-level Event-TimeRAF design remains insufficient.

II. LITERATURE REVIEW AND RELATED WORK

A. Time-Series Forecasting

Time-series forecasting has traditionally been studied through statistical models, machine-learning regressors, and deep neural networks. Autoregressive and seasonal models remain useful baselines because they are interpretable and efficient. Gradient-boosted tree models such as XGBoost and LightGBM are widely used for tabular forecasting with lagged and rolling features [8], [9]. Recurrent models such as LSTM and GRU can capture sequential dependence [10], [11]. More recently, Transformer-based forecasting models have used patching and channel-independent designs to handle long lookback windows more efficiently [12]. For air-quality forecasting, these approaches are helpful, but they may still be sensitive to distribution shifts and unobserved external events.

B. Time-Series Foundation Models

TSMs aim to provide transferable forecasting ability across datasets and domains. TimesFM uses a decoder-only architecture for time-series forecasting [2]. Moirai proposes a universal forecasting Transformer trained on a large-scale open time-series archive [3]. MOMENT provides a family of open time-series foundation models for general-purpose time-series tasks [4]. Chronos converts time-series values into tokens and trains language-model architectures for probabilistic forecasting [5]. TTM proposes compact pre-trained models for efficient zero/few-shot forecasting [6]. These models motivate the use of pre-trained forecasting backbones, but practical domain adaptation is still difficult when the downstream series responds strongly to local events and environmental conditions.

C. Retrieval-Augmented Forecasting

Retrieval-augmented generation first became popular in NLP by combining parametric models with non-parametric memory [13]. Dense Passage Retrieval (DPR) introduced dense dual-encoder retrieval for open-domain question answering [14]. TimeRAF adapts the retrieval-augmentation idea to zero-shot time-series forecasting [7]. It constructs a time-series knowledge base, retrieves top- k candidate windows using a learnable retriever, and combines them through Channel Prompting. The base paper reports TTM-Base as the main backbone, context length 512, forecast horizon 96 in most experiments, and $k = 8$ as the default number of retrieved candidates. Its ablations compare random retrieval, cosine retrieval, learnable retrieval, alternative integration mechanisms, candidate counts, knowledge-base choices, and backbone choices.

Event-TimeRAF follows the TimeRAF principle but changes both the knowledge source and the target domain.

Instead of relying only on numerical time-series retrieval, it uses a hybrid knowledge base containing historical PM2.5 windows, weather summaries, calendar context, and event evidence. The implemented system is TimeRAF-inspired rather than a full reproduction: it evaluates random retrieval, cosine retrieval, event-aware hybrid retrieval, drift-aware feature fusion, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt forecast fusion, while a learnable dual-encoder retriever and full Channel-Prompting neural integration remain future extensions.

D. Air-Quality Forecasting and Data Sources

PM2.5 forecasting often uses historical pollutant measurements together with meteorological factors such as temperature, humidity, wind speed, precipitation, and pressure. The US Environmental Protection Agency publishes quality-controlled hourly observations through AirData and the Air Quality System [15]. NOAA’s Integrated Surface Database provides hourly meteorological observations from surface stations [16]. NOAA also publishes structured Storm Events records and historical Hazard Mapping System fire and smoke products [17], [18]. These official sources support a source-preserving Los Angeles County forecasting pipeline without requiring manually authored event records.

E. Explainable and Event-Aware Forecasting

Explainability is important in environmental forecasting because users need to understand why a model predicts a rise or fall in pollution. SHAP values provide a model-agnostic framework for estimating feature contributions [19]. However, explanations based only on feature importance may miss temporal and contextual cues. Event-TimeRAF therefore generates explanations from structured evidence: recent PM2.5 persistence, weather flags, retrieved similar historical windows, retrieved event records, calendar context, and drift indicators. The explanation module uses templates grounded in stored evidence, not unrestricted language-model generation.

III. PROBLEM FORMULATION

Let x_t denote the PM2.5 measurement at time t . For a forecast time t , the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t], \quad (1)$$

where L is the lookback length. Let W_t denote weather covariates aligned with the lookback window, C_t denote calendar features, and E_t denote event context retrieved from an event/news knowledge base. The target is the future PM2.5 sequence

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \dots, x_{t+H}], \quad (2)$$

where H is the forecasting horizon.

The forecasting goal is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, R_t^{ts}, R_t^{ev}, D_t), \quad (3)$$

where R_t^{ts} is retrieved time-series knowledge, R_t^{ev} is retrieved event knowledge, and D_t contains concept-drift indicators.

TABLE I
MAIN SYMBOLS USED IN THE PROPOSED FORMULATION

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event/news knowledge
D_t	Drift indicators
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

The implemented target setting is $L = 168$ hours and $H = 24$ hours.

The evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \quad (4)$$

where the earliest 70% of observations are used for training, the next 15% for validation, and the latest 15% for testing. Random splitting is not used because it would leak future temporal information into training.

Table I summarizes the essential notation.

IV. METHODOLOGY

A. Overview

Fig. 1 shows the Event-TimeRAF pipeline. The first step is to align air-quality observations with weather, calendar, and availability-filtered event features. The system then creates sliding windows and a time-series knowledge base. The implemented hybrid retriever ranks historical windows using PM2.5, weather, calendar, and event-context similarity. A separate source-record lookup links recent event identifiers to explanations; it does not supply future event labels to the model. The retrieved evidence is combined with structured features to generate a forecast, uncertainty proxy, drift score, and explanation.

B. Relationship to TimeRAF

TimeRAF retrieves top- k candidate time-series windows from a curated knowledge base and incorporates the retrieved candidates using Channel Prompting [7]. It uses a learnable dual-encoder retriever, dot-product similarity, a frozen TSFM backbone, and candidate augmentation during retriever training. Event-TimeRAF keeps the retrieval-augmentation principle, but applies it to Los Angeles County PM2.5 forecasting and expands the knowledge sources beyond numerical sequences.

The implemented system is not a complete clone of TimeRAF. Instead, it implements TimeRAF-style retrieval baselines using random retrieval, cosine similarity, and a transparent hybrid similarity score. A dual-encoder MLP retriever, candidate augmentation, and full TSFM-based Channel Prompting are reserved for later extensions. The final run includes a frozen Chronos-Bolt gate and a validation-weighted forecast-level fusion between Chronos and retrieved historical forecasts.

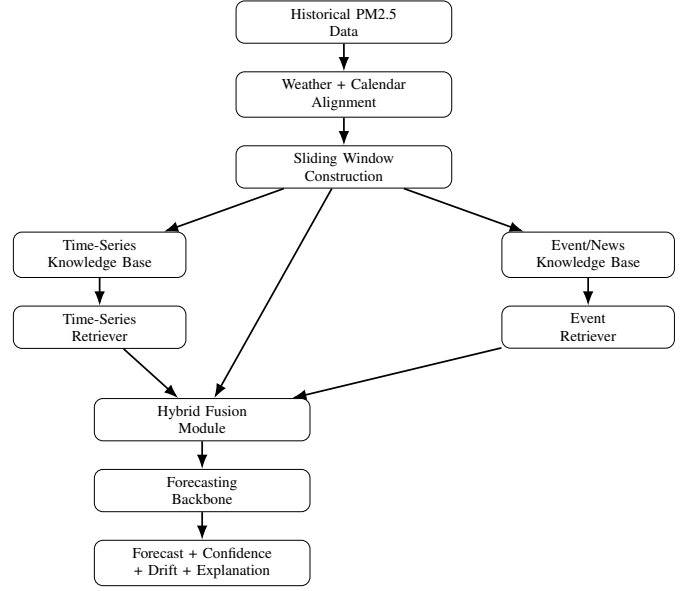


Fig. 1. Event-TimeRAF architecture used by the implemented pipeline.

C. Dataset Construction

Hourly PM2.5 observations are taken from US EPA Air-Data/AQS parameter 88101 for Los Angeles County [15]. The data audit selects one monitoring site using coverage and continuity criteria when a single site satisfies the coverage gate. In the final run, no single site is used as the target; the audited fallback is the hourly county median across official Los Angeles County AQS monitors. Timestamps are converted to *America/Los_Angeles* with explicit daylight-saving handling. Short input gaps may be filled only from past-available observations; targets are never interpolated. Weather covariates come from the nearest suitable NOAA Integrated Surface Database station [16]. Calendar features include hour, weekday, weekend, month, season, US federal holidays, and California holidays.

All normalization statistics are fitted on the training split only. The same normalization procedure is used for query windows and retrieved candidate windows, following the TimeRAF requirement that input and retrieved sequences use consistent preprocessing.

D. Time-Series Knowledge Base

The time-series knowledge base contains historical PM2.5 windows. Each record stores a window identifier, start time, end time, input window values, future window values, weather summary, calendar summary, normalized window vector, data split, and normalization identifier. Knowledge-base origins are separated by 192 hours, equal to the 168-hour input and 24-hour future combined. For every query, a candidate is eligible only when its complete future ends before the query input begins. Validation and test retrieval is additionally restricted to training-history windows. These rules exclude self-matches, near-duplicate windows, and overlap with the query context.

Given a query window X_t , the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \dots, r_{t,k}^{ts}\}, \quad (5)$$

where k is the number of retrieved candidates. The default setting is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$.

E. Event Knowledge Base

The event knowledge base contains source-preserving records relevant to Los Angeles County. NOAA Storm Events is the required structured source, while NOAA Hazard Mapping System fire and smoke records remain an optional enrichment [17], [18]. Stored fields include event identifier, event start and end, information-availability assumption, location, category, source, source URL, title, and summary. Event categories include wildfire, smoke, excessive heat, high wind, heavy rain, flood, dust, and other weather-related records.

Model inputs represent events with hourly publication-assumption counts, rolling 24-hour, 72-hour, and 168-hour counts, category-specific 72-hour counts, active-event counts, and an event-burst ratio. Recent source event identifiers are retained only as explanation evidence. Every event used as a model input must be available by the forecast origin under a recorded availability rule. Evaluation distinguishes events published in the prior 72 hours, events active at the origin, and recorded event intervals overlapping the target horizon. Target overlap is used only as a post-hoc label and never as a model input. NOAA Storm Events does not provide a machine-readable publication timestamp; consequently, the implementation records event start as a retrospective availability assumption. Results that use this assumption are labeled sensitivity experiments rather than strict operational forecasts. Strict experiments require source records with genuine publication or issuance timestamps.

F. Hybrid Retrieval Score

The hybrid retrieval score combines time-series similarity, weather relevance, calendar relevance, and event relevance:

$$s(q, i) = \alpha s_{ts}(q, i) + \beta s_w(q, i) + \gamma s_c(q, i) + \delta s_e(q, i), \quad (6)$$

where s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity, and s_e is event relevance. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$.

G. Fusion and Forecasting Backbone

The feature-level fusion method combines PM2.5 lag/rolling features, weather features, calendar features, retrieved time-series summary features, event features, and drift indicators. The tabular forecasting backbone uses 24 direct XGBoost regressors, one for each forecast horizon, because this design is efficient on Kaggle and provides a strong baseline. The foundation-model gate additionally evaluates frozen Chronos-Bolt and a validation-weighted forecast-level fusion between Chronos and retrieved historical forecasts. A later extension may use an MLP residual fusion module inspired by Channel Prompting:

$$z_i = \text{Concat}(h_t, r_{t,i}), \quad (7)$$

$$h_t^* = h_t + \frac{1}{k} \sum_{i=1}^k \text{MLP}(z_i), \quad (8)$$

TABLE II
METHODOLOGICAL COMPARISON BETWEEN TIMERAF AND
EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF
Target domain	General zero-shot TSF	Los Angeles County PM2.5 forecasting
Knowledge base	Time-series windows	PM2.5 windows, weather, calendar, events
Retriever	Learnable dual encoder	Random/cosine first; learnable later
Integration	Channel Prompting	Feature fusion; forecast-level TSFM retrieval fusion
Backbone	Frozen TSFM	Direct XGBoost; frozen Chronos-Bolt gate
Explanations	Retrieved-case visualization	Evidence-based forecast explanation
Drift handling	Not central	Explicit drift indicators

where h_t is the input representation and $r_{t,i}$ is the representation of the i -th retrieved candidate. This mirrors the TimeRAF idea of preserving the original input representation while adding uniformly averaged information extracted from retrieved candidates.

H. Concept Drift Detection

The implementation represents operational distribution shift using transparent indicators rather than claiming perfect concept-drift identification. The indicators are recent mean shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. A simple drift score can be defined as

$$D_t = \lambda_1 z_{\mu,t} + \lambda_2 z_{\sigma,t} + \lambda_3 (1 - \bar{s}_{ts,t}) + \lambda_4 z_{w,t} + \lambda_5 b_{event,t}, \quad (9)$$

where $z_{\mu,t}$ and $z_{\sigma,t}$ are normalized mean and variance shifts, $\bar{s}_{ts,t}$ is average retrieval similarity, $z_{w,t}$ is weather-pattern shift, and $b_{event,t}$ is an event-burst score. The implemented components are robustly scaled against training data and averaged with equal weight; the alert threshold is selected from validation scores.

I. Explanation Generation

The explanation module generates evidence-based explanations. It uses signed local XGBoost contributions averaged across the 24 direct horizon models, weather flags, retrieved historical windows, retrieved event records, calendar context, forecast direction, component-level drift evidence, and an uncertainty proxy combining retrieved-trajectory spread with validation residual error. It does not generate explanations from an unrestricted language model. Each explanation is traceable to stored machine-readable evidence.

Table II summarizes the methodological relation between TimeRAF and Event-TimeRAF.

TABLE III
FINAL DATA-READINESS AUDIT

Gate	Value	Status
PM2.5 observed coverage	0.9997	Pass
Weather complete coverage	0.9942	Pass
Event days	117	Pass
Event overlap days	2,192	Pass
Qualifying event categories	3	Pass
Strict event availability	False	Caveat

V. EXPERIMENTAL SETUP

This section reports the executed final Kaggle experiment. The run identifier is 20260723T112033170131Z. All model-selection choices were frozen before reading the final test results, and the results notebook rejected non-final or mixed-run artifacts.

A. Data Sources and Audit

The study period is 2019–2024. PM2.5 observations were taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [15]. No single monitoring site was used as the final target; the audited target is the hourly county median across official Los Angeles County AQS monitors. This fall-back selected 06–037–COUNTY, with 52,602 observed target hours and observed PM2.5 coverage of 0.9997. The PM2.5 unit was consistently recorded as micrograms per cubic meter.

Weather covariates were taken from NOAA Global Hourly/ISD through the official NOAA Open Data Dissemination public buckets [16]. The selected station was 72297023129, Long Beach / Daugherty Field / Airport, 10.28 km from the PM2.5 target centroid. The complete weather-feature coverage after allowed past-only short-gap filling was 0.9942, and the raw measured core coverage was 0.9905.

Environmental events were loaded from hash-verified NOAA Storm Events annual detail archives [17]. Because Kaggle could not reliably download NCEI Storm Events directly, the experiment used a private delivery cache containing unchanged NOAA annual archives and a source manifest with official URLs, byte counts, and SHA-256 hashes. The audit retained the delivery mode `verified_official_noaa_cache`. The event audit found 117 event days, 2,192 source-overlap days, and three qualifying event categories. NOAA Storm Events does not expose machine-readable publication timestamps; therefore, event-aware outputs use the recorded event start time as a retrospective availability assumption. They are not strict real-time event forecasts.

B. Forecasting Task and Split

The forecasting task predicts PM2.5 from one to 24 hours ahead using a 168-hour lookback window. The split is chronological: the earliest 70% of valid windows are used for training, the next 15% for validation, and the latest 15% for testing. The final test set contains 7,199 forecast origins and 172,776 forecasted hourly target points. Random splitting is not used.

C. Models

The evaluated models are persistence, daily seasonal naive, weekly seasonal naive, a direct XGBoost model using PM2.5 features, a direct XGBoost context model using PM2.5, weather, and calendar features, random retrieval, cosine retrieval, XGBoost with cosine retrieval features, Event-TimeRAF without drift features, full Event-TimeRAF with event and drift features, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt forecast fusion. The Chronos checkpoint is `amazon/chronos-bolt-small`; it completed successfully in the final run, so the foundation-model publication gate is satisfied.

D. Metrics and Statistical Checks

The primary metrics are MSE, MAE, RMSE, MAPE, sMAPE, and R^2 . MSE is retained for comparability with the TimeRAF base paper, while MAE and RMSE provide more interpretable PM2.5 error scales. Subset metrics are reported only when at least 50 forecast origins are available. The event subset contains 532 origins, the active-event subset contains 293 origins, and the drift subset contains 560 origins, so all pre-specified subsets pass the minimum-size gate. Pairwise comparisons use paired block bootstrap intervals with 24-hour blocks and 500 resamples.

VI. RESULTS AND DISCUSSION

Table IV reports the final test-set results. The strongest overall model is the XGBoost context baseline, which uses PM2.5 history, weather, and calendar features. It obtains MSE 26.185 and MAE 3.125. The full Event-TimeRAF XGBoost variant obtains MSE 26.712 and MAE 3.149. Therefore, this run does not support a claim that the full Event-TimeRAF XGBoost variant outperforms the strongest non-retrieval context baseline on the all-origin test set. It does, however, substantially outperform the persistence and seasonal naive baselines.

The frozen Chronos-Bolt publication gate completed successfully. The validation fusion search selected a retrieval-fusion weight of 0.75. On the test set, retrieval-augmented Chronos reduces MSE from 28.941 to 28.709, an improvement of about 0.8% relative to frozen Chronos-Bolt. The paired block-bootstrap comparison gives an MSE difference of -0.233 with a 95% interval from -0.665 to 0.152, so the direction is favorable but not conclusive under this uncertainty estimate. MAE changes from 3.205 to 3.209, which is practically flat.

Table V summarizes the main ablation comparisons. The clearest supported gain is from adding weather and calendar context to the PM2.5-only XGBoost model: MSE decreases by 3.764 and the confidence interval is fully below zero. Cosine retrieval improves over random retrieval in the retrieval-only setting. In contrast, adding cosine retrieval features to the already strong XGBoost context model does not improve the all-origin result in this run. Event-hybrid retrieval and drift indicators also do not show a clear overall improvement beyond the context baseline.

The k sensitivity results in Table VI show that the retrieval-only cosine baseline improves as more neighbors are used

TABLE IV
FINAL OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

Model	MSE	MAE	RMSE	R^2
M00 persistence	47.649	4.137	6.903	-0.130
M01 daily seasonal	48.706	4.057	6.979	-0.155
M02 weekly seasonal	65.606	5.123	8.100	-0.556
M03 XGBoost PM2.5	29.949	3.368	5.473	0.290
M04 XGBoost context	26.185	3.125	5.117	0.379
M05 random retrieval	42.367	4.257	6.509	-0.005
M06 cosine retrieval	38.083	3.940	6.171	0.097
M07 XGBoost cosine	26.672	3.151	5.164	0.367
M08 Event-TimeRAF no drift	26.677	3.145	5.165	0.367
M09 Event-TimeRAF full	26.712	3.149	5.168	0.367
M10 frozen Chronos-Bolt	28.941	3.205	5.380	0.314
M11 Chronos + retrieval	28.709	3.209	5.358	0.319

TABLE V
SELECTED PAIRED BLOCK-BOOTSTRAP ABLATION RESULTS

Comparison	MSE diff.	95% CI low	95% CI high
M04 minus M03, weather/calendar	-3.764	-5.856	-2.071
M06 minus M05, cosine/random retrieval	-4.284	-5.862	-2.759
M07 minus M04, retrieval features	0.487	0.015	1.028
M08 minus M07, event hybrid	0.005	-0.292	0.275
M09 minus M08, drift indicators	0.035	-0.101	0.161
M09 minus M04, full Event-TimeRAF	0.528	-0.026	1.182
M11 minus M10, Chronos retrieval	-0.233	-0.665	0.152

TABLE VI
COSINE RETRIEVAL SENSITIVITY TO k

k	MSE	MAE	RMSE
1	65.129	5.133	8.070
4	41.237	4.134	6.422
8	38.083	3.940	6.171
16	36.488	3.851	6.040

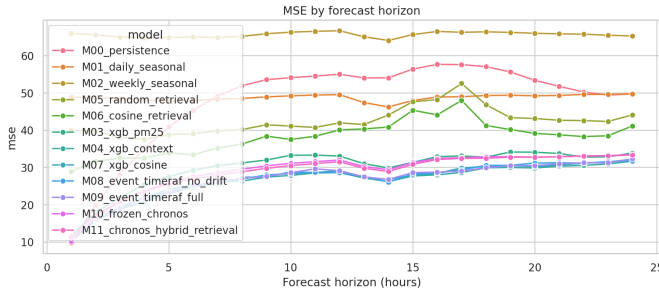


Fig. 2. MSE by forecast horizon for the final test run.

within the tested range. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is $k = 16$, with MSE 36.488. The final Event-TimeRAF feature models used the pre-specified default $k = 8$.

Fig. 2 and Fig. 3 show horizon-wise test errors. Fig. 4 summarizes retrieval behavior, and Fig. 5 shows the drift-score distribution across the test period. These plots are generated directly from the saved final-run artifacts and share run identifier 20260723T112033170131Z.

The main interpretation is that conventional supervised context features are very strong for this Los Angeles County 24-

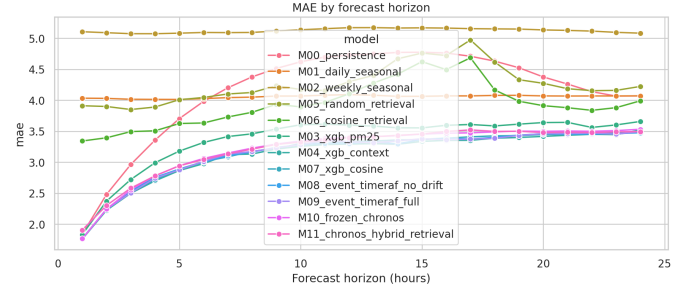


Fig. 3. MAE by forecast horizon for the final test run.

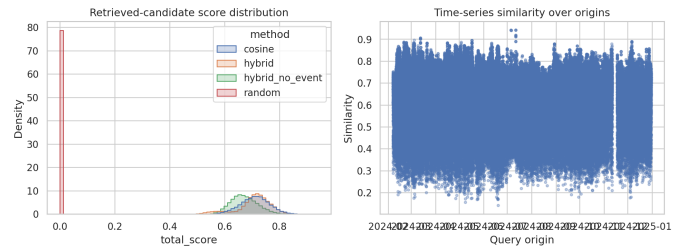


Fig. 4. Retrieval diagnostics for the final test run.

hour PM2.5 task. Retrieval is useful as a standalone historical-pattern baseline and gives a small favorable MSE change when fused with frozen Chronos-Bolt, but the present feature-level Event-TimeRAF design does not beat the best XGBoost context baseline overall. This negative result is important for the final paper: Event-TimeRAF should be presented as an implemented and audited event-aware retrieval framework with mixed empirical effects, not as a uniformly superior forecasting model.

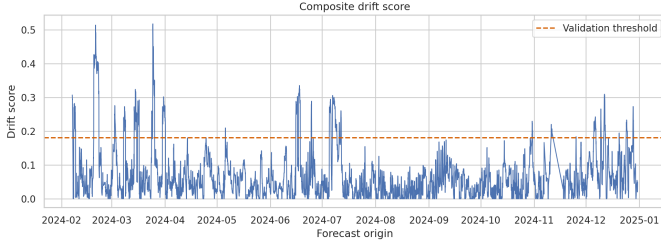


Fig. 5. Drift-score diagnostics for the final test run.

Finally, all event-aware interpretations must be read with the availability caveat. NOAA Storm Events records are official and hash-verified, but the source does not provide machine-readable publication timestamps. The experiment records event start as the availability time, so event-related results are retrospective sensitivity results. A strict operational study would require event or smoke sources with genuine issue or publication times, such as an audited NOAA HMS cache or alert product archive.

VII. CONCLUSION

This paper presented Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Los Angeles County PM_{2.5} forecasting under distribution shift. The implemented pipeline combines official EPA PM_{2.5} observations, NOAA weather, hash-verified NOAA Storm Events records, leakage-safe time-series retrieval, direct XGBoost forecasters, drift indicators, evidence-based explanations, and a frozen Chronos-Bolt publication gate.

The final 2019–2024 Kaggle experiment shows that the strongest overall model is the XGBoost weather-calendar context baseline, not the full Event-TimeRAF variant. The full Event-TimeRAF XGBoost model remains competitive with the context baseline but does not improve the all-origin test score. The frozen Chronos-Bolt gate completes successfully, and retrieval-augmented Chronos gives a small favorable MSE change over frozen Chronos, although the bootstrap interval does not establish a conclusive improvement. These findings suggest that retrieval and event context are promising but need stronger integration than the current feature-level design to reliably outperform strong tabular context models.

The main limitation is event availability. NOAA Storm Events records are official and source-verified, but they lack machine-readable publication timestamps. As a result, the event-aware results are retrospective event-start sensitivity results rather than strict operational event forecasts. Future work should add event sources with genuine issue times, improve event-category representation, test learnable retrievers, and evaluate deeper foundation-model integration closer to TimeRAF’s Channel Prompting design.

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